TRIMMER ASSEMBLY LINE

Monte Carlo · Yamazumi · Flow Simulation Report

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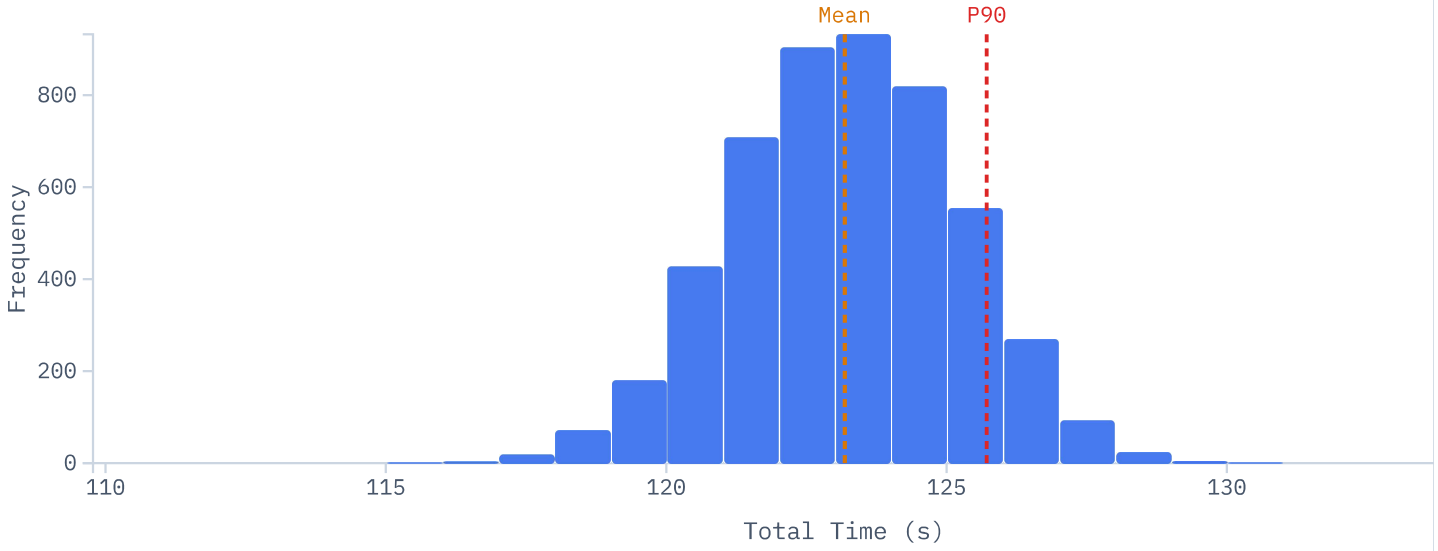
EXPECTED (MEAN)	MINIMUM
123.18 seconds	115.55 seconds

MAXIMUM	STD DEVIATION
130.07 seconds	1.99 ± seconds

DISTRIBUTION — TOTAL CYCLE TIME



Frequency · Mean (123.2s) · P90 (125.7s)

VALUE-ADDED ANALYSIS

VAA: 123.2s (100.0%) **Semi-VA:** 0.0s (0.0%) **NVA:** 0.0s (0.0%)

SHIFT CAPACITY FORECAST & CUSTOMER DEMAND

UPDATE PLAN

AVAILABLE WORKING TIME
8 hrs 0 mins (28,800s)

REQUIRED TAKT TIME (PER UNIT)
23.11 s

EXPECTED DAILY OUTPUT (MEAN FORECAST)
1,299 units

P90 GUARANTEE (CONSERVATIVE)
1,246 units

P10 POTENTIAL (OPTIMISTIC)
1,339 units

PERCENTILE BREAKDOWN (EXPECTED OUTPUT)

PCT	OUTPUT (UNITS)	CONF.	VISUAL
P10	**1246**	10%	
P25	**1284**	25%	
P50	**1307**	50%	
P75	**1324**	75%	
P80	**1328**	80%	
P85	**1333**	85%	
P90	**1339**	90%	
P95	**1348**	95%	
P99	**1361**	99%	

TASK CONTRIBUTION

TASK	TYPE	AVG (S)	% OF TOTAL	STN
1. tán_co_terminal	VAA	21.76	17.7%	1
2. Housing_top_lò_xo	VAA	13.53	11.0%	2
I/O automatic machine	VAA	11.75	9.5%	3
4. hipot,lazer,_test_dây	VAA	12.36	10.0%	4
5. xịt,_chèn,_đầu_lưỡi,_blade_guard	VAA	17.94	14.6%	5
6. hdpe,_hộp_nhựa,_phụ_kiện	VAA	21.64	17.6%	6
7. color_box,_keo	VAA	11.52	9.3%	7
8. master_carton,_keo	VAA	12.70	10.3%	8

TARGET DAILY OUTPUT ANALYSIS

PROBABILITY OF PRODUCING 450 UNITS/DAY

100.0%

≥ TARGET (MET)

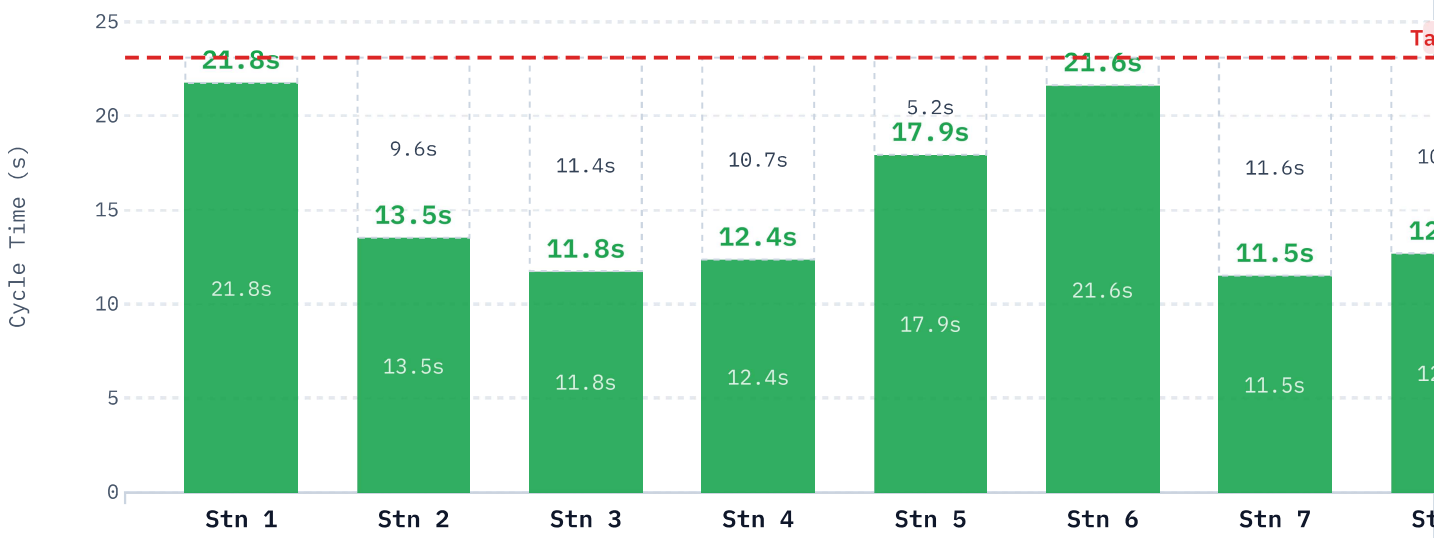
100.0%

< TARGET (MISSED)

0.0%

YAMAZUMI — OPERATOR LOADING CHART

Takt Time (s) 23.1



VAA (Green) — 100.0% Semi-VA (Yellow) — 0.0% NVA (Red) — 0.0% Unbusy / Idle

STATIONS	PEAK STATION
8	Stn 1
operators	21.76s

BALANCE EFFICIENCY	OVER TAKT
70.8%	0
line balance	stations / 8

FLOW SIMULATION — BOTTLENECK & UTILIZATION ANALYSIS

500 units simulated (DES)

BOTTLENECK
Stn 1
21.77s / unit

THROUGHPUT RATE
165
units / hour

LINE EFFICIENCY
70.8%
balance rate

AVG SYSTEM WAIT
3.89s
queue delay / unit

STATION	TASKS ASSIGNED	AVG CYCLE (S)	UTILIZATION	AVG QUEUE WAIT	AVG QUEUE DEPTH	STATUS
Stn 1	1. tán_co_terminal	21.77	99.0%	0.00s	0.00	BOTTLENECK
Stn 2	2. Housing_top_lò_xo	13.58	61.7%	0.00s	0.00	OK
Stn 3	I/O automatic machine	11.75	53.4%	0.00s	0.00	OK
Stn 4	4. hipot,lazer,_test_dây	12.34	56.1%	0.00s	0.00	OK
Stn 5	5. xịt,_chèn,_dầu_lưỡi,_blade_guard	18.04	82.0%	0.00s	0.00	HIGH LOAD
Stn 6	6. hdpe,_hộp_nhựa,_phụ_kiện	21.65	98.5%	3.88s	0.80	HIGH LOAD
Stn 7	7. color_box,_keo	11.50	52.3%	0.00s	0.00	OK
Stn 8	8. master_carton,_keo	12.67	57.6%	0.00s	0.00	OK